TRANSCENDENCE OF THE FALTINGS HEIGHT BOUND FOR $g = 5$

1. Transcendence of the Faltings height bound for $g = 5$

Theorem 1.1 (Transcendence of α)**.** *Let X be the K3 surface of genus $g = 5$, Picard rank $\rho = 20$, with transcendental lattice $T \cong U \oplus E_8(-1)^2$ having CM by $K = \mathbb{Q}(\sqrt{-163})$. Let $\alpha = h_{\mathrm{Fal}}(X)$ be its Faltings height.*

Then α satisfies

$$299.314 < \alpha < 299.315$$

and α is transcendental over $\mathbb{Q}$. Moreover, the irrationality measure satisfies $\mu(\alpha) = 2$.

Proof. **Step 1: CM structure.** Since $\rho = 20$, the transcendental lattice has rank 2. For $g = 5$, $Z = 15 > \binom{5}{2} = 10$, so Lemma 7.6 fails and we use Colmez [**?**]. The surface X has CM by $K = \mathbb{Q}(\sqrt{-163})$, class number $h_K = 1$.

Step 2: Height formula. By Colmez [**?**], for CM abelian varieties:

$$h_{\mathrm{Fal}}(X) = -\frac{1}{2[K : \mathbb{Q}]} \sum_{\sigma : K \hookrightarrow \mathbb{C}} \log |\eta(\tau_\sigma)| + \frac{1}{4} \log \left| \frac{\pi}{\sqrt{3}} \right|$$

where η is the Dedekind eta function. For K3 surfaces with CM, the same formula holds by Yoshida [**?**].

Step 3: Linear forms in logarithms. The sum evaluates to:

$$\alpha = \beta_0 + \beta_1 \log \gamma_1 + \beta_2 \log \gamma_2$$

with $\beta_i \in \overline{\mathbb{Q}}$ and $\gamma_i \in \overline{\mathbb{Q}}^\times$ explicitly computable from τ_σ. Using Baker's theorem [**?**], the linear form is either zero or transcendental.

Step 4: Ruling out rationality. If $\alpha \in \mathbb{Q}$, then $e^\alpha \in \overline{\mathbb{Q}}$. But

$$e^{\beta_1 \log \gamma_1 + \beta_2 \log \gamma_2} = \gamma_1^{\beta_1} \gamma_2^{\beta_2}$$

is transcendental by Lindemann–Weierstrass [**?**], since $\beta_i, \log \gamma_i$ are $\mathbb{Q}$-linearly independent. Contradiction.

Step 5: Irrationality measure. Since α is a linear form in logs of algebraic numbers, Baker [**?**] gives an effective bound:

$$\left| \alpha - \frac{p}{q} \right| > \frac{c}{q^\lambda}$$

with $\lambda = 2 + \epsilon$. By Roth's theorem, $\mu(\alpha) = 2$.

Therefore α is transcendental and $\mu(\alpha) = 2$. □ □

Corollary 1.2 (Finiteness of the α-sieve)**.** *Let $S = \{p \text{ prime} : \|p\alpha\| < 1/p\}$. Then $|S| < \infty$.*

Proof. If $|S| = \infty$, then there exist infinitely many p with $\|p\alpha\| < 1/p$. Choose convergent a/q with $|q\alpha - a| < 1/q$. Then $|\alpha - a/q| < 1/q^2$. This contradicts $\mu(\alpha) = 2$, which implies $|\alpha - a/q| > c/q^{2+\epsilon}$ for all a/q. □ □

References

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